1)

**Ethics:** Ethics are fundamental principles of right and wrong that guide a person's behaviour. They are often subjective and based on philosophical or cultural values.

Example:

A developer discovers a major or minor privacy flaw in an app. While not illegal to ignore it, they feel an ethical obligation to report it because it's the right thing to do for the users.

**Laws:** Laws are formal rules created by government authorities. They are enforceable by the state, and breaking them results in specific penalties.

Example: Under the GDPR or Digital Security Act, a company is legally required to notify users of a data breach within a specific timeframe. Failing to do so results in a massive fine.

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**Professional Responsibility**: This is the specific set of duties and standards expected of people in a particular profession. These are often defined by bodies like ACM or IEEE.

Example: An engineer refuses to sign off on a software release because the testing phase was skipped. Even if no law was broken, they are exercising professional responsibility to ensure public safety and product quality.

02)

**Principles and Professional Responsibility**: Software engineering ethics go beyond technical skill; they require engineers to ensure their work is safe, reliable, and beneficial to society. Key responsibilities include:

- Pubic Interest: Prioritizing the safety and well being of the public above all else.
- Confidentially: Protecting client data and intellectual property.
- Integrity: Remaining honest and avoiding conflicts of interest or "cutting corners" in testing.

The ACM/IEEE Code of Ethics: The ACM/IEEE Code serves as a standard for professional behavior and framework for resolving ethical dilemmas. It guides decision-making through eight key principals.

- Public: Act in public interest
- Client: Act in their best interest
- Product: Ensure the highest professional standards.

- Judgment : Maintain Integrity and independence
- Management : Ethical approach to software management

## Comparison between Utilitarianism and Deontology :

| Feature | Utilitarianism | Deontology |
|---|---|---|
| Core Idea | The outcome determines if an action is right | The action itself must follow a moral rule/duty |
| Goal | Greatest good for the greatest number | Do your duty, regardless of consequences |
| Focus | Minimizing harm and maximizing utility | Categorical Imperative |

**Virtue Ethics:** Virtue ethics focuses on the character of the person rather than specific rules (Deontology) or consequences.

**Guiding Ethical Decision-Making:** Virtue Ethics guides decisions by aiming for the "Golden Mean".

Example:

- **Testing:** Deficiency: Being lazy/negligent (no testing).
- **Excess:** Being obsessive (testing forever, never releasing).
- **Virtuous Mean:** Diligent testing that ensures safety while meeting reasonable deadlines.

**Key Principles of ACM Code:** The association for computing machinery (ACM) code is organized into four main sections, but the core principles for writing are:

1. General Ethical Principles: contribute to society and human well-being; avoid harm; be honest and trustworthy; be fair and take action not to discriminate

2. Professional Responsibilities: strive for the highest quality in processes and products.

3. Professional Leadership Principles: Manage resources to enhance the quality of working life; articulate, encourage acceptance of, and evaluate fulfillment of social responsibilities.

4. compliance with the Code: Uphold, promote and respect the principles of the code.

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## Importance of ACM Codes:

- **Public Trust:** They assure the public that the profession is committed to safety and ethics, not just profit.

- **Standardization:** they establish a uniform set of "rules of engagement" across the global tech community.

- **Professional Identity:** They define what it means to be a "professional" rather than just a "coder", raising the status of the field.

## Whistleblowing:

Whistleblowing is the act of an employee or insider disclosing wrongdoing, illegal activity or unethical behavior within an organization to an internal authority or an external

(like the media or government)

## Ethical considerations:

- Duty to the public (primary): Most codes (ACM/IEEE) state that the public's safety and well-being override all other obligations. If a system flaw could kill or harm people, whistleblowing is an ethical must.

- Duty to the Employer: Professionals have a duty of loyalty and confidentiality to their company.

## Challenges:

- Retaliation & Job Loss: Whistleblowers often face firing, being "blacklisted" in the industry.

Legal Battles: Companies may sue the professional for breach of contract or "theft" of trade secrets.

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